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Tapped, Not Filled: A Field Audit of Hydration-Station Activations Against the Living Dashboard's Bottles-Saved Tally

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Abstract. Slop University's twenty-two campus hydration stations report to the School of Continuous Improvement's Living Dashboard through a single channel: each station's infrared activation sensor increments a shared counter, converted at a fixed per-activation coefficient into the dashboard's headline "bottles saved" tile. We report a covert field audit pairing shift-based direct observation at fourteen stations with a campus-wide self-report intercept survey, cross-referenced against six weeks of raw counter logs. Trained observers coded a stratified sample of 512 activations into one of four categories — full fill, partial fill, rinse, or false trigger — and 238 surveyed users independently estimated their own weekly refill frequency. Full fills account for well under half of logged activations at every station audited, and the survey-implied weekly fill total sits well below what the counter's raw activation count would suggest. A duration-based ablation excluding activations under three seconds lowers the term's cumulative tally by roughly two-fifths without changing its placement in the University's Sustainability Tile funding tier at any of six weekly checkpoints. We discuss what a headline conservation figure gains, and forfeits, when it counts a sensor trigger rather than a bottle.

Introduction

Slop University operates twenty-two hydration stations across campus — infrared-triggered bottle-filling points installed in three phases since 2023 as part of the University's single-use-plastic reduction commitment. Each station's activation sensor reports to a shared telemetry feed maintained by Facilities, which the School of Continuous Improvement's Living Dashboard converts into a running total on its Sustainability Tile: the "bottles saved" figure quoted in the annual report, on orientation signage, and in the funding submission that sets Facilities' discretionary Green Impact allocation for the following term. The conversion is arithmetically simple — one activation, one bottle saved — and administratively convenient: the sensor already existed for maintenance scheduling, and repurposing its count as an impact figure required no new instrumentation.

What the conversion assumes, and does not check, is that an activation is a fill. A station's infrared beam breaks whenever anything crosses it for long enough to register: a completed fill, a partial top-up abandoned mid-flow, a hand rinsed under the spout, or a passer-by testing whether the light comes on. Facilities' own maintenance documentation treats the sensor as reliable for its original purpose, flagging a filter change at a fixed activation threshold; nothing in that documentation, or in the dashboard specification that inherited the same count, distinguishes a fill from anything else that trips the beam.

This paper reports a covert field audit of the hydration-station telemetry feed against what actually happens at the tap. We make three contributions, each hedged appropriately:

- We report a shift-based direct-observation audit coding 512 activations across fourteen

stations into fill, partial-fill, rinse, and false-trigger categories, and compare the resulting cause distribution against station traffic.
- We report an independent self-report intercept survey estimating aggregate weekly fills, and compare its implied total against the raw counter log for the same stations and weeks.
- We evaluate a duration-based ablation of the counter log and show it moves the Sustainability Tile's headline total substantially without moving the funding tier the total is used to set.

The remainder of the paper situates the audit against sensor-validation and self-report-discordance literature, describes the observation and survey instruments, reports the cause breakdown, the survey-counter comparison, and the ablation, and discusses what a conservation figure gains, and forfeits, when it counts a sensor trigger rather than a bottle.

Related work

Campus water-bottle refill stations are a well-studied conservation intervention: post-installation audits report substantial reductions in disposable-bottle purchasing and disposal [1], and adoption studies outside North America report comparable uptake once stations sit convenient to daily routes [2]. Neither literature audits what the stations' own sensors count as an activation. The passive-infrared and related occupancy-class sensors underlying most refill-station and building-automation telemetry are well documented to register non-target events — airflow, transient motion, incidental proximity — alongside the behaviour they are nominally installed to detect [3], and controlled field comparisons against ground-truth logs find accuracy dropping precisely in the low-motion, low-duration regime a brief rinse or test tap occupies [4].

A parallel body of work compares an automated activity count against independently observed or self-reported ground truth. Electronic hand-hygiene monitoring, structurally the closest analogue to a hydration-station counter, correlates only partially with direct human observation and diverges further once observers are no longer visibly present [5], [6]. Self-reported physical activity is comparably unreliable against device-measured ground truth, generally over-estimating volume relative to an accelerometer [7], [8], [9], [10], a pattern consistent with a broader tendency to over-report behaviour a respondent expects to be viewed favourably, including pro-environmental conduct specifically [11]. The public-management literature on composite and headline performance indicators has separately argued that a measure adopted as a reportable target tends to acquire effects on the institution's behaviour independent of, and sometimes instead of, the outcome it was built to track [12], [13], a dynamic formalised most directly by Goodhart's law and its variants [14], [15].

Method

Twenty-two hydration stations operate across Slop University's campus, each fitted with the infrared activation sensor Facilities installed for filter-change scheduling and later repurposed as the Living Dashboard's Sustainability Tile input. The Tile converts the raw weekly activation count directly into the term's headline "bottles saved" figure:

$$\text{BottlesSaved}_t = c \cdot \sum_{s=1}^{22} A_{s,t}$$

where $A_{s,t}$ is station s's logged activation count in week t and $c = 1$ is the credit Facilities assigns per activation. The resulting cumulative total sets the term's Green Impact funding tier: Tier 1 below 4,000 cumulative units, Tier 2 from 4,000 to 65,000, and Tier 3 above 65,000.

We audited fourteen of the twenty-two stations across a six-week window, stratified into eight high-traffic and six low-traffic stations by prior-term logged volume. Two trained observers worked unobtrusive three-hour shifts — seated with a laptop or reading nearby rather than positioned as a visible monitoring presence, on the grounds that even reanalysis debunking the dramatic textbook account of observer effects still finds subtler, measurable responsiveness to being watched in the underlying data [16] — coding each activation's cause from four categories: full fill, partial fill (flow interrupted before the bottle appeared full), rinse (hands or an open container held briefly under the spout), and false trigger (no container present, including deliberate tests of the sensor). Fifteen percent of shifts were independently double-coded to establish inter-rater agreement, computed as Cohen's kappa following standard guidance for nominal observational categories [17]; agreement on the double-coded subsample was substantial ($\kappa = 0.81$).

Coding yielded 512 events: 316 at high-traffic stations, 196 at low-traffic stations.

Independently, we intercepted 238 station users campus-wide across the same six weeks and asked each to estimate their own typical weekly refill frequency at that station, aggregated to a per-station weekly mean. Self-report measures of this kind are known to diverge from device-logged ground truth in adjacent behavioural domains, generally over-estimating relative to a sensor [7], and self-reported pro-environmental behaviour specifically has been shown to fall once individual responses are no longer attributable [11]; we treat the survey as an independent, imperfect estimate rather than as ground truth in its own right, and compare it against the raw counter log for the fourteen audited stations.

Each sensor's firmware records activation duration alongside the count. As an ablation, we recomputed the weekly tally excluding activations under three seconds — a threshold set from the coded sample's own duration distribution, where full fills registered a median 9.4 seconds against 1.8 seconds for rinses and false triggers combined — and compare the resulting cumulative total, and its funding-tier placement, against the unfiltered figure Facilities currently reports.

Results

Direct observation finds full fills a minority cause of activation at every station audited (Figure 1). At high-traffic stations, 46% of coded activations were full fills, 21% partial fills, 17% rinses, and 16% false triggers; at low-traffic stations the full-fill share falls further, to 29%, with rinses and false triggers together accounting for 45% of coded activity. The pattern held across the six-week window: no single week's coded sample at either traffic tier moved the full-fill share by more than four percentage points from its window average.

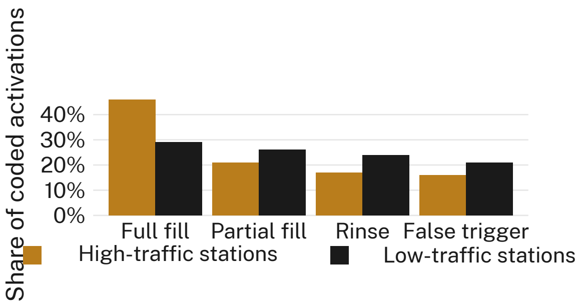

Figure 1: Direct observation of 512 activations at fourteen stations: full fills account for 46% of activity at high-traffic stations and 29% at low-traffic stations, where rinses and false triggers together outnumber fills.

The survey-counter comparison finds only a weak association between what a station's sensor logs and what its users report doing there (Figure 2). Across the fourteen audited stations, mean weekly counter-logged activations predict mean weekly survey-implied fills at $r = 0.31$ ($n = 14$); several of the highest-logging stations sit in the middle of the self-reported distribution, and one of the lowest-logging stations reports among the highest self-reported fill frequency in the sample.

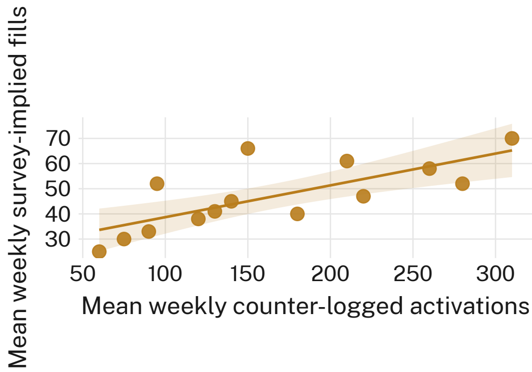

Figure 2: Mean weekly counter-logged activations predict mean weekly survey-implied fills only weakly across the fourteen audited stations ($r = 0.31$, $n = 14$); logged volume and reported use rank stations differently.

The duration ablation moves the headline total substantially without moving the figure it feeds (Figure 3). Excluding activations under three seconds lowers the term's cumulative tally from 51,800 to 32,300 units by week six — a 38% reduction — while leaving the Sustainability Tile's Tier 2 funding placement unchanged at all six weekly checkpoints (Tier 2 spans 4,000-65,000 cumulative units; both series remain inside that band throughout the window). Facilities retains the unfiltered count for continuity with the total reported in prior terms.

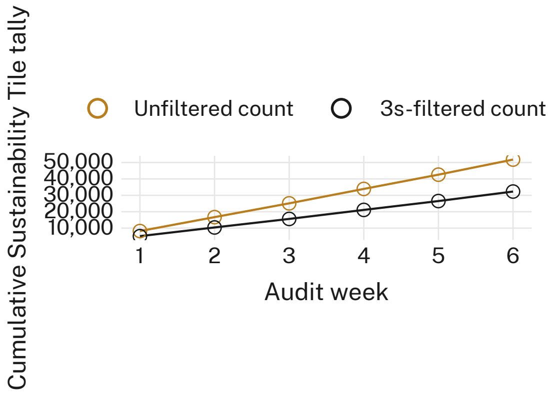

Figure 3: Excluding activations under three seconds lowers the term's cumulative Sustainability Tile tally from 51,800 to 32,300 units by week six, a 38% reduction that leaves the Tier 2 funding placement unchanged at all six weekly checkpoints.

Read together, the three results describe a headline figure whose composition observation does not support, whose relationship to independently reported behaviour is weak, and whose sensitivity to a defensible measurement choice does not reach the threshold that would change its administrative consequence.

Discussion

The pattern here echoes a broader observation in headline sustainability reporting: a figure can move, and be reported as moving, without the underlying practice it purports to track moving with it. Comparable divergence between a disclosed sustainability metric and independently assessed substance has been documented at organisational scale in ESG reporting, where automated content analysis finds disclosed indicators and independently coded practice diverging systematically enough to warrant purpose-built detection methods [18]. The hydration-station case is smaller and, we think, more mundane than the ESG literature's disclosure-gaming concern: nothing in this audit suggests Facilities deliberately selected a favourable count. The sensor was installed for filter scheduling, the dashboard specification adopted its output because the output already existed, and the resulting figure inherited the sensor's blind spots — rinse, test, partial fill — without anyone treating that inheritance as a measurement decision. A proxy adopted for convenience carries the same downstream weight as one adopted deliberately once it is load-bearing for a funding tier [12]. Whether the same divergence holds at the eight stations outside this audit's scope, we do not know.

Limitations

Our audit covers fourteen of twenty-two stations across one six-week window at a single institution, though the consistency of the cause breakdown across both high- and low-traffic strata suggests the pattern is not confined to the stations sampled. The three-second duration threshold is drawn from this study's own coded distribution rather than an external standard, though the funding-tier result holds under the unfiltered count as well, which required no threshold choice at all. Covert observation avoids inflating fill behaviour via observer presence but cannot rule out some seasonal or timetable-linked variation in station use outside the audited weeks. The self-report survey's 238 respondents are a convenience intercept sample rather than a stratified draw of the campus population.

Conclusion

Slop University's hydration-station "bottles saved" figure counts a sensor activation, and direct observation finds a completed fill responsible for under half of what that sensor logs at every station audited, falling further at low-traffic stations. An independent self-report estimate correlates only weakly with the same counter log, and a defensible duration-based correction to the count moves the headline total by more than a third without moving the funding tier the total sets. None of this implies the hydration stations are not saving bottles; it implies the number attached to how many is closer to a count of beam-breaks than a count of fills. Facilities, and any unit reporting a sensor-derived conservation figure elsewhere on campus, might usefully audit what the sensor is actually counting before the count sets a budget line. Future work might instrument the fill event directly, at the tap, rather than at the door.

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